

**SRN: PES1UG24CS456**

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**Section: H**

## **Problem Statement #27**

### **Warehouse Inventory & Pallet Location Tracker**

#### **Problem Context**

An automated warehouse management system tracks 3D shelf pallet positions, enforces maximum shelf/rack weight limits, and logs stock movements through barcode/RFID scanning.

#### **Primary Stakeholders / Actors**

1. Warehouse Operator
2. Logistics Supervisor
3. Barcode/RFID Scanner

The first two actors are explicitly specified in the problem statement. The scanner is included as an external device that interacts directly with the system for barcode/RFID-based stock movement logging.

| <b>Req ID</b> | <b>Type</b> | <b>Description</b>                                                                                                                | <b>Priority</b> | <b>Acceptance Criteria</b>                                                                                                                                                                                 | <b>Rationale</b>                            |
|---------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>FR-001</b> | Functional  | The system shall validate proposed pallet placement coordinates against the rack maximum load capacity before confirming storage. | High            | <b>Pass:</b> Pallet is assigned only when the selected rack has sufficient remaining capacity.<br><b>Fail:</b> Placement is rejected when the pallet would cause the rack's rated capacity to be exceeded. | Prevents unsafe or overloaded rack storage. |

|               |            |                                                                                                                                                                                    |        |                                                                                                                                                                                                 |                                                                         |
|---------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>FR-002</b> | Functional | The system shall allow a Warehouse Operator to record a pallet's SKU, quantity, and 3D shelf location using barcode/RFID scanning.                                                 | High   | <b>Pass:</b> After a valid scan, the SKU, quantity, and shelf coordinates are recorded successfully. <b>Fail:</b> The system does not create a stock record when the scan is invalid.           | Provides accurate and efficient inventory recording.                    |
| <b>FR-003</b> | Functional | The system shall allow authorized users to search for a pallet using its SKU and display its current 3D shelf location.                                                            | High   | <b>Pass:</b> Entering a valid SKU returns the pallet's current shelf, row, and level coordinates. <b>Fail:</b> An invalid or unavailable SKU returns an appropriate "Pallet Not Found" message. | Enables warehouse personnel to locate inventory quickly.                |
| <b>FR-004</b> | Functional | The system shall record every confirmed pallet placement, removal, or movement with the pallet identifier, previous location, new location, timestamp, and user/device identifier. | Medium | <b>Pass:</b> A completed stock movement creates a log containing all required fields. <b>Fail:</b> A movement is not considered complete if its corresponding log entry cannot be created.      | Maintains traceability of inventory movements.                          |
| <b>FR-005</b> | Functional | The system shall allow a Logistics Supervisor to view current rack occupancy and pallet weight information for warehouse inventory.                                                | Medium | <b>Pass:</b> The supervisor can select a rack and view its occupied capacity and stored pallet information. <b>Fail:</b> The system does not display incomplete or                              | Helps supervisors monitor warehouse capacity and inventory utilization. |

|                |                                        |                                                                                                               |      |                                                                                                                                                                                                                           |                                                               |
|----------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
|                |                                        |                                                                                                               |      | unavailable rack data as valid information.                                                                                                                                                                               |                                                               |
| <b>NFR-001</b> | Nonfunctional — Performance & Security | The warehouse item lookup service shall locate any SKU pallet coordinate across 100,000 bins in under 100 ms. | High | <p><b>Pass:</b> Benchmarking confirms that SKU lookup completes in less than 100 ms across a dataset containing 100,000 bins under the specified test load.</p> <p><b>Fail:</b> The lookup exceeds 100 ms.</p>            | Ensures rapid inventory location during warehouse operations. |
| <b>NFR-002</b> | Nonfunctional — Reliability            | The system shall preserve all confirmed stock-movement records without loss during normal system operation.   | High | <p><b>Pass:</b> After at least 1,000 consecutive valid movement transactions, all 1,000 corresponding movement records can be retrieved from the log. <b>Fail:</b> Any confirmed transaction is missing or corrupted.</p> | Ensures reliable inventory history and traceability.          |

## 2. ACTORS

### Actor 1 — Warehouse Operator

The Warehouse Operator performs day-to-day inventory operations such as scanning pallets, placing pallets in racks, moving pallets, and searching for inventory.

### Actor 2 — Logistics Supervisor

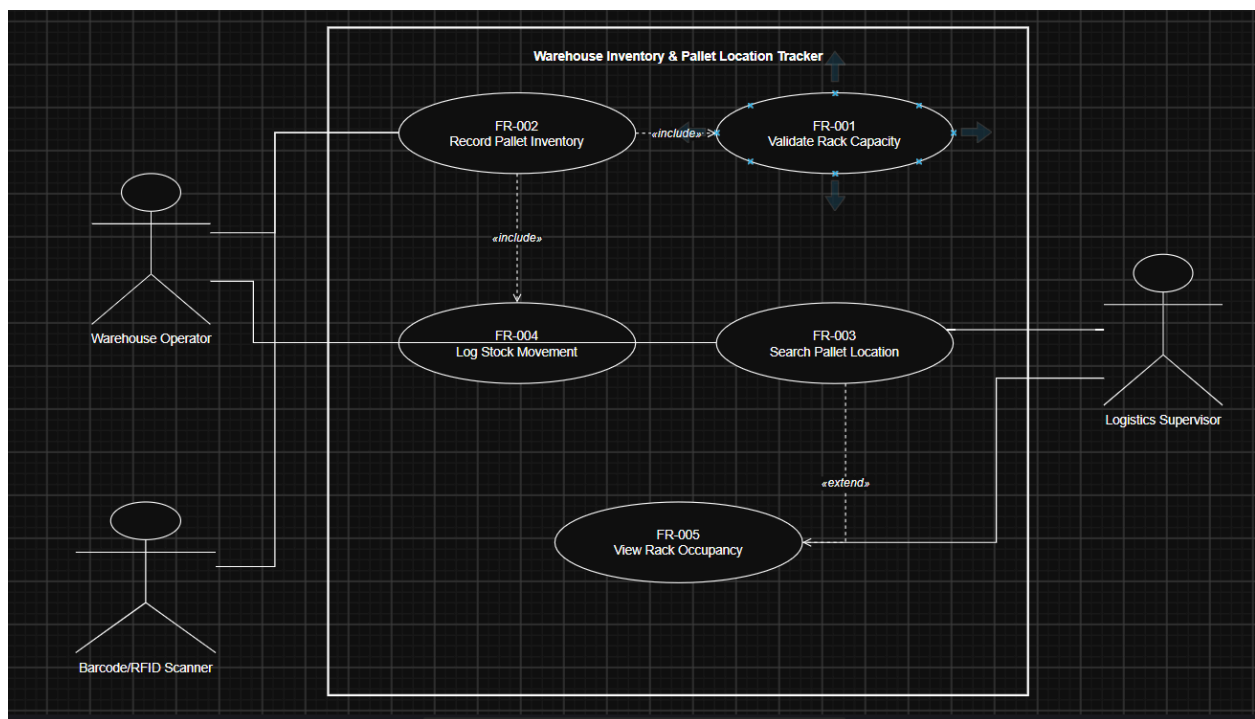
The Logistics Supervisor monitors warehouse inventory, rack occupancy, pallet weights, and movement information.

### Actor 3 — Barcode/RFID Scanner

The Barcode/RFID Scanner provides pallet identification information to the system when inventory is scanned during storage, removal, or movement.

Absolutely. This is the **complete Use-Case Flow Specification** you can paste directly into your document. It follows the exact structure required by your problem statement: **preconditions**, **postconditions**, **main success scenario**, and **alternate flow**.

## UML Diagram



## Use-Case Flow Specification

### UC-02 — Record Pallet Inventory

#### Primary Actor

Warehouse Operator

## Supporting Actor

Barcode/RFID Scanner

## Brief Description

This use case describes how a Warehouse Operator records a pallet's SKU, quantity, and 3D shelf location using barcode/RFID scanning. Before confirming the placement, the system validates the destination rack's maximum load capacity and logs the confirmed stock movement.

## Preconditions

1. The Warehouse Operator is authorized to perform inventory operations.
2. The pallet has a valid barcode/RFID identifier.
3. The destination rack and shelf location exist in the warehouse system.
4. The barcode/RFID scanner and inventory system are operational.

## Postconditions

### Success:

1. The pallet's SKU, quantity, and 3D shelf location are recorded.
2. The rack occupancy information is updated.
3. The confirmed stock movement is logged with the pallet identifier, previous location, new location, timestamp, and user/device identifier.

### Failure:

1. The pallet is not assigned to the proposed location.
2. The inventory information remains unchanged.
3. The rejected movement is not recorded as a completed stock movement.

## Main Success Scenario

1. The Warehouse Operator selects **"Record Pallet Inventory."**
2. The system requests the pallet's barcode/RFID information.
3. The Warehouse Operator scans the pallet.
4. The system retrieves the pallet's SKU and existing inventory information.
5. The Warehouse Operator enters or selects the pallet quantity and destination 3D shelf coordinates.
6. The system validates the proposed pallet placement against the destination rack's maximum load capacity.
7. The system confirms that sufficient rack capacity is available.
8. The system records the pallet's SKU, quantity, and new 3D shelf location.
9. The system updates the rack occupancy information.

10. The system logs the confirmed stock movement with the required movement details.
11. The system displays **"Pallet Inventory Recorded Successfully."**
12. The use case ends successfully.

### **Alternate Flow — Rack Capacity Exceeded**

**Condition:** The proposed pallet placement would exceed the destination rack's maximum rated capacity.

1. The system determines that the destination rack does not have sufficient remaining capacity.
2. The system rejects the proposed placement.
3. The system displays **"Insufficient Rack Capacity."**
4. The Warehouse Operator is prompted to select another valid rack or shelf location.
5. If another location is selected, the system repeats the capacity validation.
6. If the operator cancels the operation, the process ends without updating the pallet location or recording the movement.